

A Century of Natural Disasters

More events, fewer deaths, bigger bills

Visual exploration of the EM-DAT Emergency Events Database, 1900-2022

● **Lakshya Jain (BT2024044)**

When? The temporal record

● **Arnav Jain (BT2024233)**

Where? The geography of exposure

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What? The impact signature of hazards

One question

Over 1900-2022, has the burden of natural disasters actually grown - and has the world become better or worse at surviving it?

A disaster has no single cost - and the four costs do not move together.

 **Counted**

15,015 events

 **Killed**

22.9M deaths

 **Displaced**

8.5B affected

 **Cost**

\$6.7T damage

Three sub-questions, one per member - so no task is split across the team.

 **Lakshya Jain**

Four things to fix before any chart

1 Semicolon-separated, comma decimals

read naively, every number arrives as text

2 2023 is a part-year

51 rows dropped, or every trend fakes a collapse

3 Trailing spaces in hazard labels

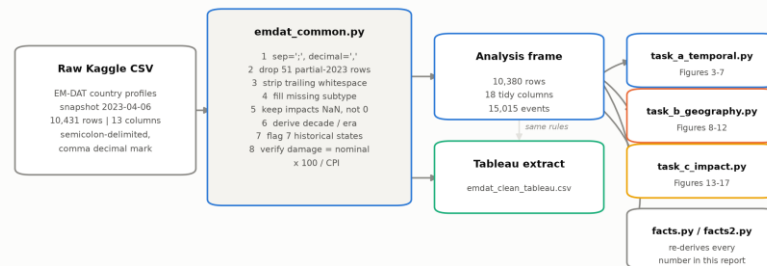
"Extreme temperature " split one category in two

4 Missing impacts kept as missing

zero-filling would flatter every historical decade

One cleaning module feeds every chart in this report

The three task sets never clean the data themselves, so all fifteen charts are guaranteed to describe an identical frame. The verification harness reads that same frame and re-derives every number quoted in this report.



Boxes are files in code/. The row counts shown are printed by the module itself on every run.

10,431 rows -> 10,380 rows · 15,015 events · 225 countries

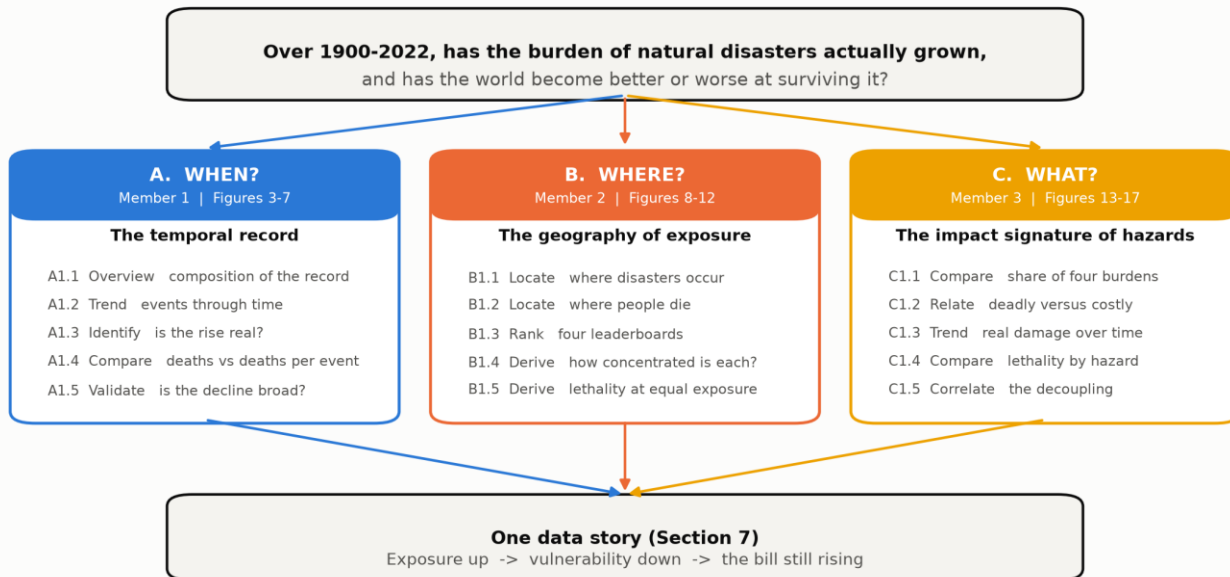
One cleaning module feeds all 15 charts, so no two figures can disagree.

 **Lakshya Jain**

One question, three owned task sets, fifteen charts

One question, three independently owned task sets, fifteen charts

Each member owns a sub-question end to end - its tasks, its figures, its inferences and its section of the report and of the video - so no task is split across the team. Task verbs follow the design space of Schulz et al.

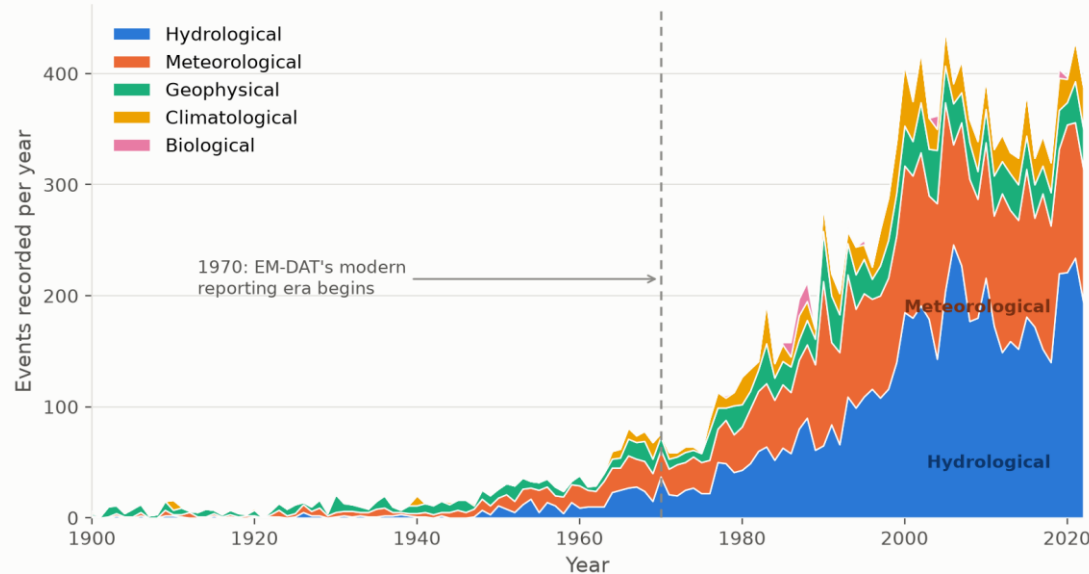


Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

The record explodes after 1970

The record explodes after 1970 - mostly floods and storms

91% of all events in the file are dated 1970-2022, a period covering only 43% of the 123 years.



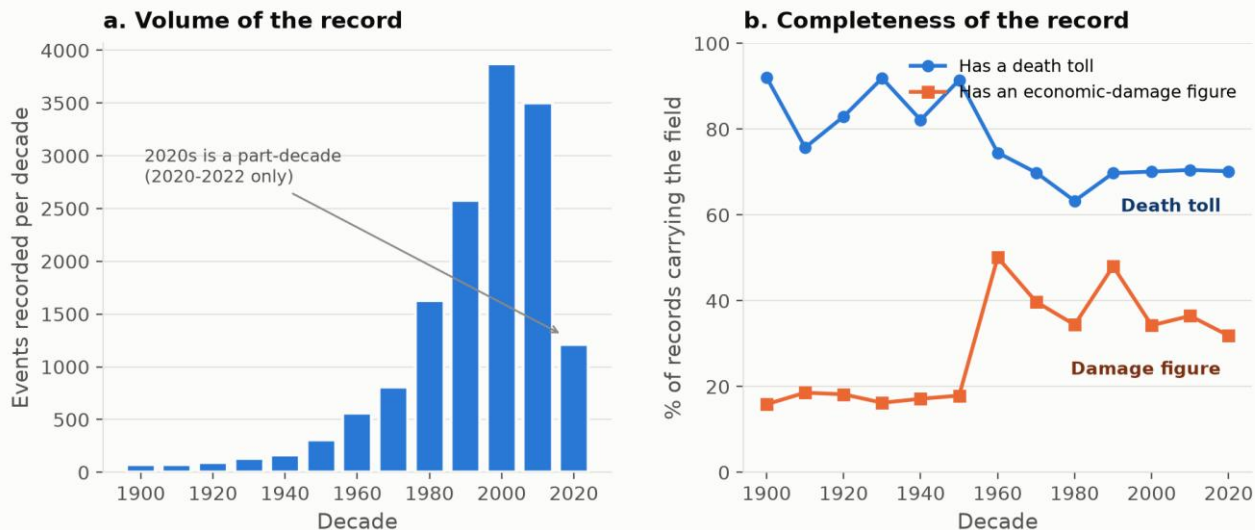
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

That shape is also exactly what a step-change in data collection looks like.

...but it is partly a count of reporting

The count of disasters is partly a count of reporting

Volume grew 52x from the 1900s to the 2000s, but the share of records carrying an economic-damage figure peaks at just 50% - so every damage total in this report is a lower bound.



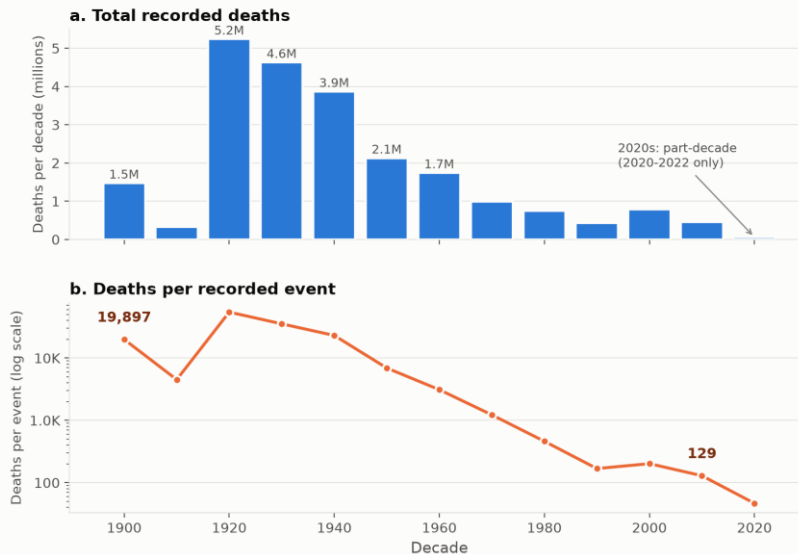
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

91% of the record sits in 43% of the years. So we use rates, not counts, from here on.

Deaths per recorded event fell 99.4%

Disasters became far more survivable even as they became far more numerous

Deaths per recorded event fell 99.4%, from ~19,900 in the 1900s to ~129 in the 2010s. The two measures use different scales, so they are drawn as two panels rather than on a twin axis.



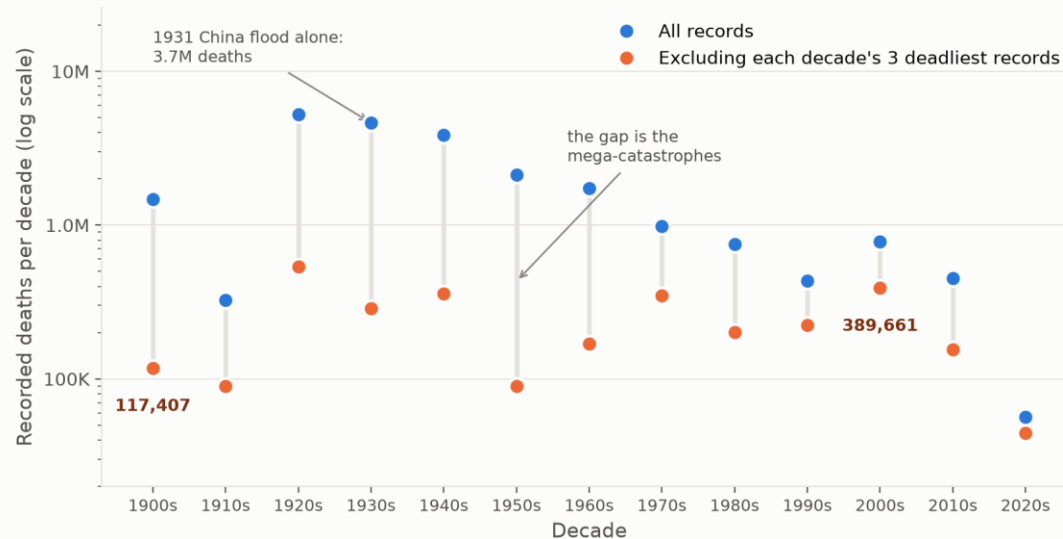
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Two panels, never a twin axis - a twin axis lets the author pick the crossing point.

Then we tried to break our own finding

The fall in deaths is the disappearance of mega-catastrophes, not a broad decline

Strip each decade's three deadliest records and the residual death toll shows no downward trend at all - the 2000s sit above the 1900s. A dot plot is used rather than bars because bar length is not proportional to value on a logarithmic axis.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

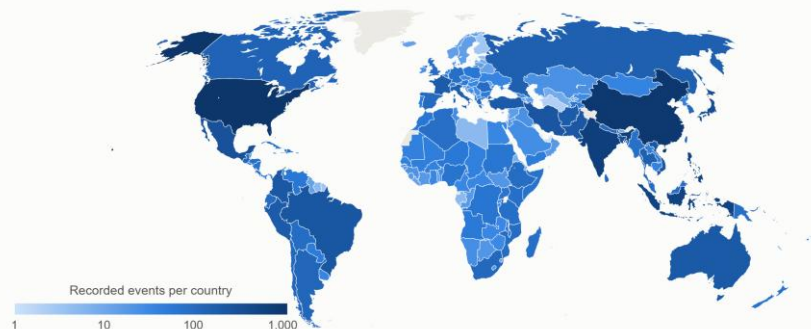
Strip each decade's three deadliest records and the decline vanishes entirely.

Disasters are global. Death is not.

Almost every country on Earth appears in the disaster record

Total recorded disaster events per country, 1900-2022.

225 countries appear; the United States (1,119) and China (984) lead.

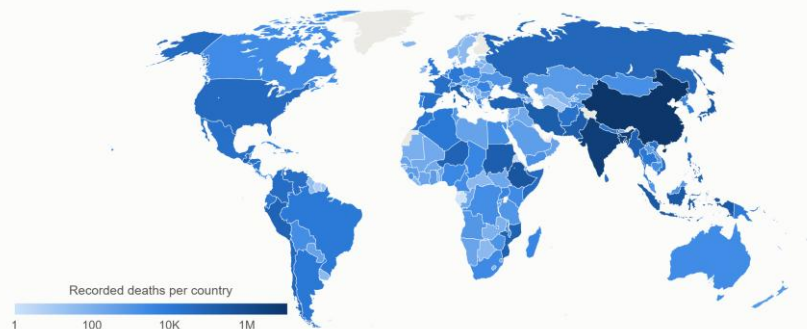


Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06. Grey = no recorded event. Colour scale is logarithmic.

But death is concentrated in a handful of countries

Total recorded disaster deaths per country, 1900-2022.

China and India alone account for more than half of all 22.9 million recorded deaths.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06. Grey = no recorded event. Colour scale is logarithmic.

All 225 countries appear

China + India = 68% of all deaths

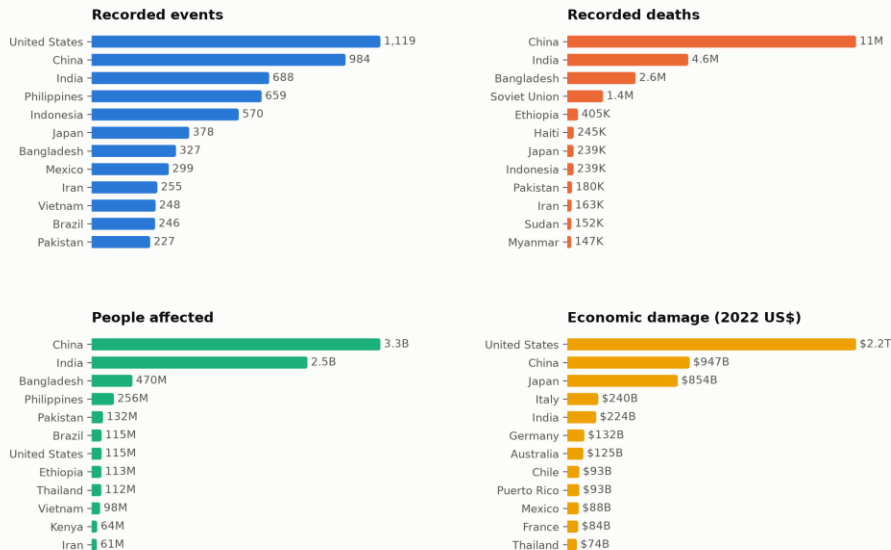
LIVE IN TABLEAU: switch the map here

Arnav Jain

Four measures, four different leaderboards

Most disasters, most deaths, most people affected and most money lost are four different maps

Top 12 countries on each measure, 1900-2022. The United States tops events and damage but ranks 23rd for deaths; Bangladesh is 3rd for deaths and 23rd for damage.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

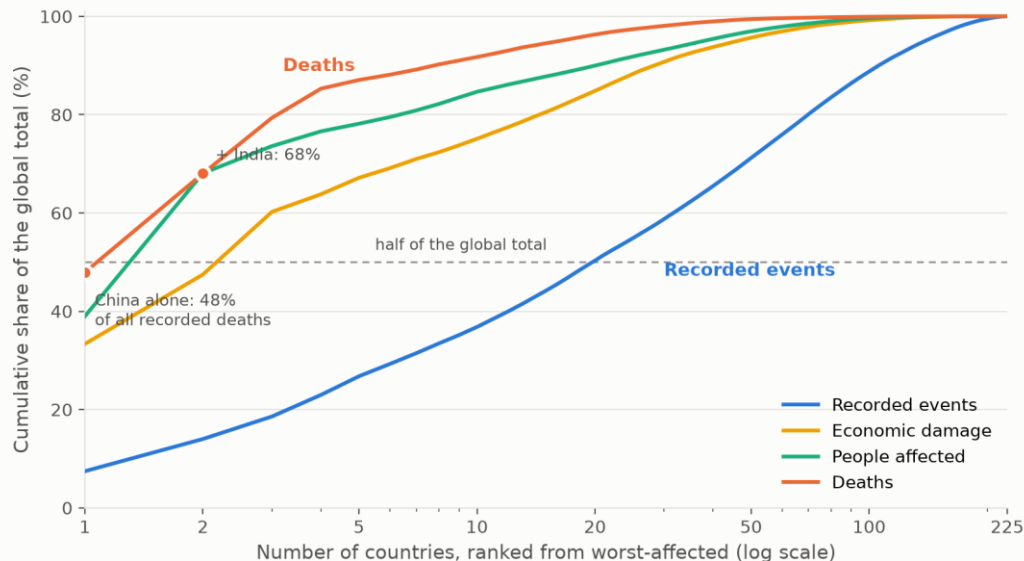
The US is 1st for events and 1st for damage - and 23rd for deaths.

Arnav Jain

Two countries hold half of all deaths

Half of all deaths fall on 2 countries; half of all disasters are spread over 20

Cumulative share of each global total, countries ranked worst-first. Deaths and people affected are the most concentrated burdens; the raw count of events is by far the most evenly spread.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

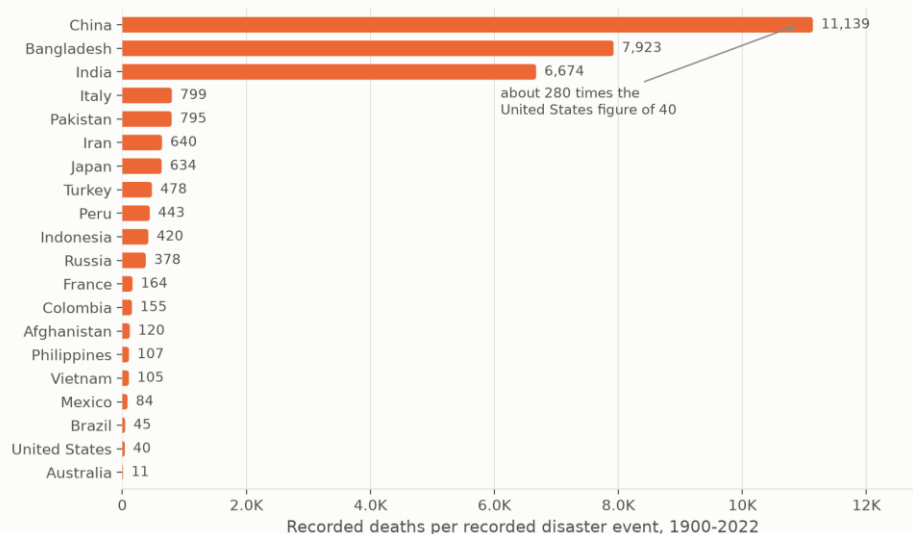
But it takes twenty countries to reach half of all recorded events.

Arnav Jain

Same exposure, 1,000x the lethality

Among equally disaster-prone countries, lethality differs by a factor of 1,000

The 20 countries with the most recorded events, ranked by deaths per event. Exposure is similar; the outcome is not - which points at vulnerability and response capacity, not at the hazards themselves.



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

China 11,139 deaths per event; Australia 11. That gap is not hazard - it is vulnerability.

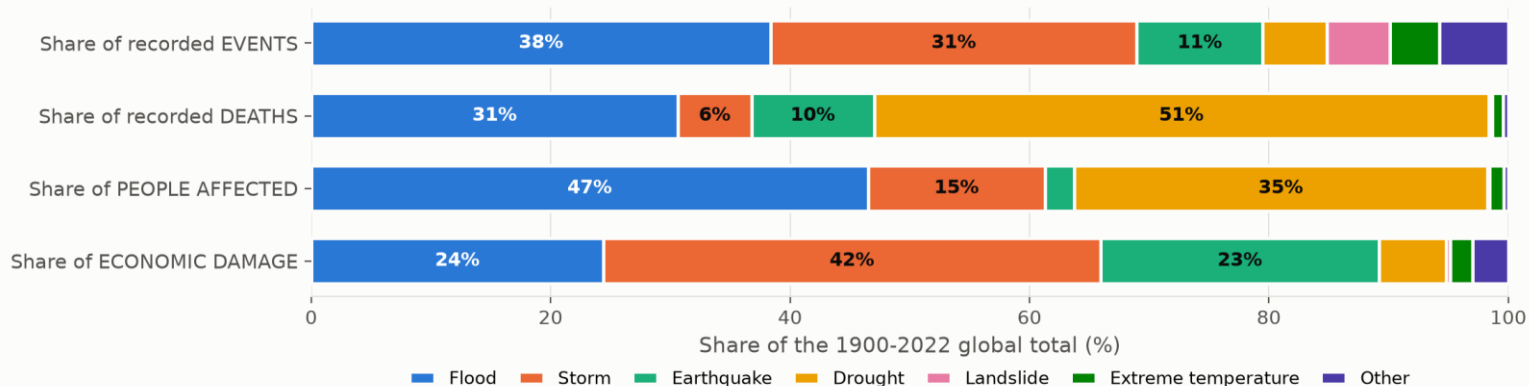
Arnav Jain

5% of events. 51% of deaths.

Four bars, the same seven hazards, the same colour order.

A hazard's share of disasters says almost nothing about its share of harm

Drought is 5% of recorded events but 51% of recorded deaths. Storms are 31% of events and 42% of damage but only 6% of deaths. The four bars use the same seven hazard categories in the same order.



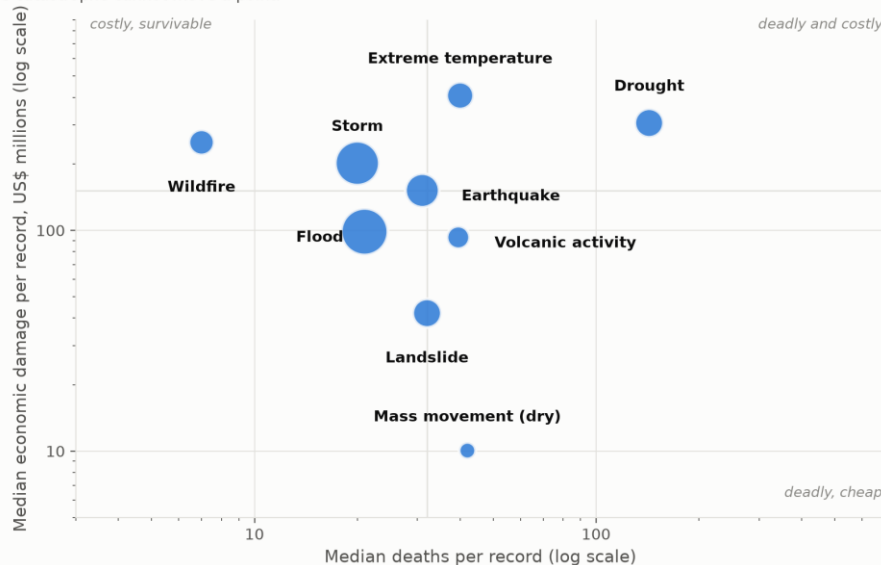
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Drought kills, storms cost, floods displace - they are different hazards.

Deadly and costly are different hazards

Drought kills on a different scale from every other hazard

Each bubble is one hazard type; bubble size is the number of recorded events. A typical drought record carries 143 deaths, more than three times the next hazard - while extreme temperature carries the largest typical damage bill (\$408M). Medians, not totals, so one catastrophe cannot move a point.



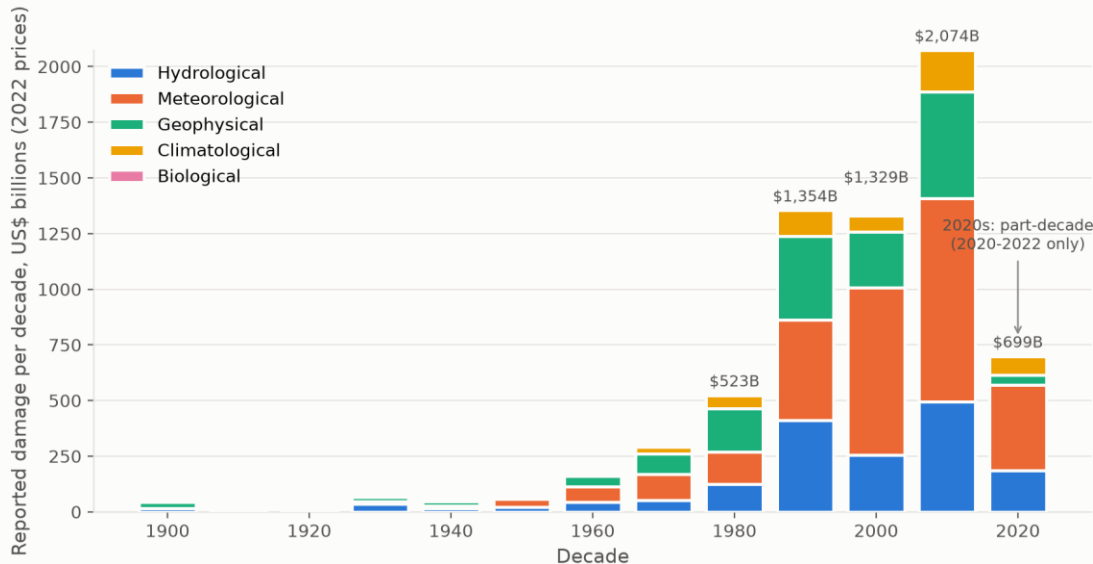
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Drought: 143 deaths per record. Wildfire: \$250M and a median of 7 deaths.

\$2.07 trillion - the costliest decade yet

The money cost of disasters keeps climbing, in real terms

Reported damage is inflation-adjusted to 2022 US dollars. Only 37% of records carry a damage figure, so these bars are lower bounds. Coverage does not improve after 1970 (it stays in a 32-48% band), so the rise cannot be an artefact of better reporting.



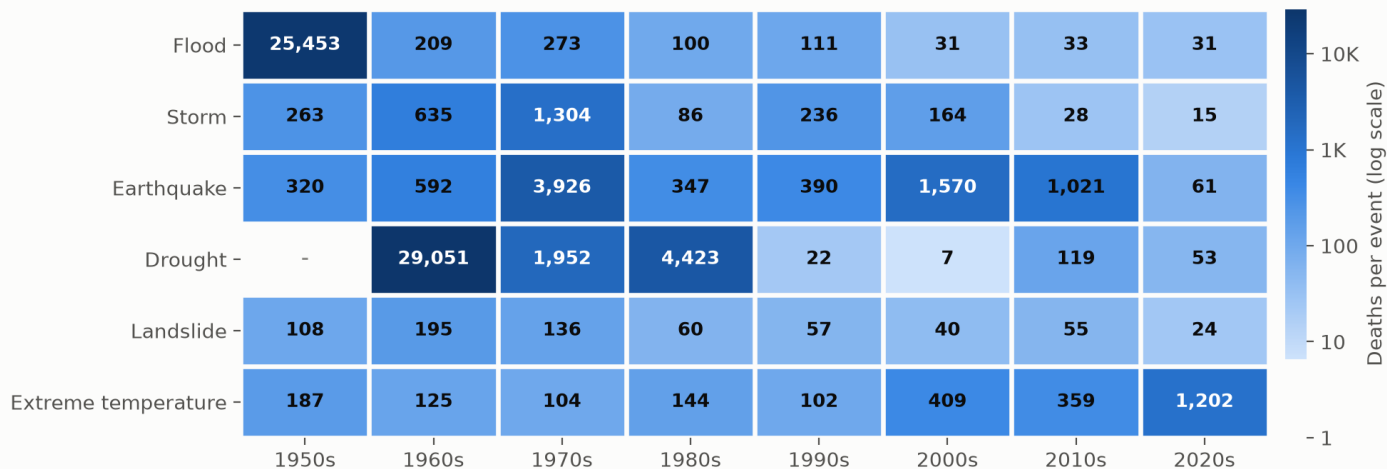
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Damage coverage does not improve after 1970, so this rise is not a reporting artefact.

Which hazards became survivable?

Famine and flood became survivable; heatwaves went the other way

Deaths per recorded event, by hazard and decade. Drought fell from 29,051 per event in the 1960s to 53 in the 2020s; extreme temperature rose from 125 in the 1960s to 1,202 in the 2020s.



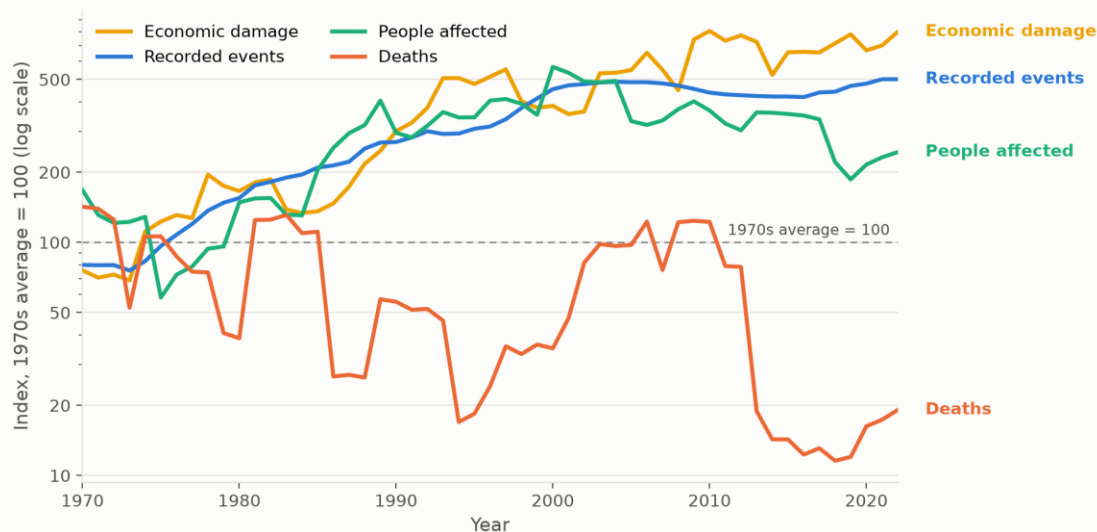
Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Drought 29,051 -> 53. Earthquakes never improved. Extreme heat went the wrong way.

Costs rose. Deaths did not follow.

Costs rose, deaths did not follow

Five-year moving averages, each indexed to its own 1970s mean so four different units share one axis. Damage grew 1.58x per decade ($p < 0.001$); annual deaths show no significant trend ($p = 0.064$), and the rank correlation between the two is -0.01 .



Source: EM-DAT Emergency Events Database country profiles (CRED / UCLouvain), 1900-2022 snapshot of 2023-04-06.

Rank correlation between annual deaths and annual damage: -0.01 .

Exposure up. Vulnerability down. The bill still rising.

 **Exposure rose** more assets, more people - and much better reporting

 **Vulnerability fell** deaths per recorded event down 99.4% in a century

 **But unevenly** 2 countries carry half of all deaths; heat is getting worse

And one caveat on our own headline: most of the fall in total deaths is the loss of mega-catastrophes, not a broad decline.

Thank you.